Project Design Phase – II

Solution Requirements (functional & non-functional)

DATE:	**27/09/2026**
TEAM ID:	**SWTID-2026-2642**
PROJECT NAME:	**Weather wise**
MAXIMUM MARKS:	**4 Marks**

2. Functional Requirements

Following are the functional requirements of the proposed solution

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through form; Store user details in MongoDB
FR-2	User Login	Login using registered email and password; Validate user credentials
FR-3	AI-FITTRACK Management	Create, view, update and delete AI-FITTRACKs; Store AI-FITTRACK details in MongoDB
FR-4	AI Answer	Generation Enter a question; Generate answers using Google Gemini AI
FR-5	AI-FITTRACK Search	Search AI-FITTRACKs using keywords; Retrieve relevant AI-FITTRACK information
FR-6	User Authentication	Secure protected routes using JWT; Allow access to authorized users

3.Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

FR NO.	Non-functional Requirement	Descrion
NFR-1	Usability	Simple and user-friendly interface for registration, login, AI-FITTRACK search and AI chat.
NFR-2	Security	Password encryption using bcrypt and JWT-based authentication for protected routes.
NFR-3	Reliability	Store and retrieve user and AI-FITTRACK data reliably using MongoDB.
NFR-4	Performance	Provide quick AI-FITTRACK search results and AI-generated answers through the backend API.
NFR-5	Availability	Allow users to access the application when the server, database and required AI service are available.
NFR-6	Scalability	Use modular frontend and backend components to support future feature expansion.